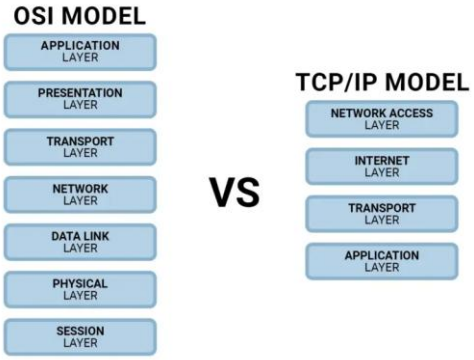


INVESTIGATING TCP/IP AND OSI MODELS IN ACTION

ABSTRACT

This lab investigates OSI and TCP/IP protocol models in a local area network using Cisco Packet Tracer. By inspecting Protocol Data Units (PDUs), the exercise demonstrates data encapsulation, parameter adaptation (MAC, IP, and ports), and the roles of core protocols like HTTP, DNS, and TCP. Tracking communications from session setup to teardown effectively bridges theoretical networking concepts with real-time operations.

Allan Ngugi

Network Security

Lab: Investigating TCP/IP and OSI Models in Action

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Introduction

This lab utilizes Packet Tracer's Simulation mode to observe and analyze the operational mechanics of the OSI and TCP/IP protocol models within a local area network. By capturing network traffic generated between a Web Client and a Web Server, the lab examines how various protocols encapsulate and exchange Protocol Data Units (PDUs) across different model layers. Through tracking events, filtering specific traffic types, and inspecting PDU headers, the exercise demonstrates how data travels through the OSI layers and illustrates the roles of core protocols like HTTP, TCP, and DNS during client-server communications.

Walkthrough and Answers to Questions

Part 1: Examine HTTP Web Traffic

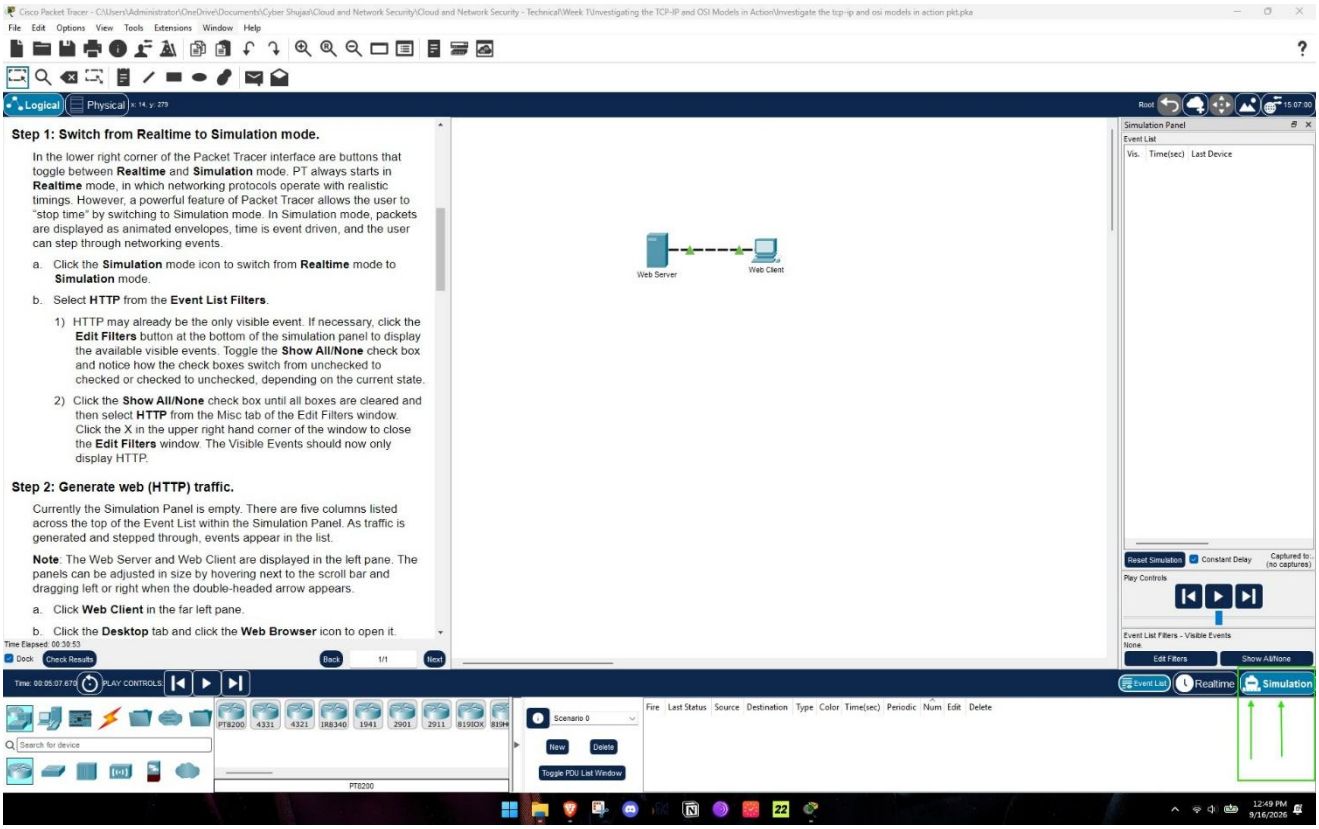
In Part 1 of this activity, you will use Packet Tracer (PT) **Simulation** mode to generate web traffic and examine HTTP.

Step 1: Switch from Realtime to Simulation mode.

In the lower right corner of the Packet Tracer interface are buttons that toggle between Realtime and **Simulation** mode. PT always starts in **Realtime** mode, in which networking protocols operate with realistic timings. However, a powerful feature of Packet Tracer allows the user to "stop time" by switching to **Simulation** mode. In **Simulation** mode, packets are displayed as animated envelopes, time is event driven, and the user can step through networking events.

a. Click the **Simulation** mode icon to switch from **Realtime** mode to **Simulation** mode.

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b. Select **HTTP** from the **Event List Filters**.

1) HTTP may already be the only visible event. If necessary, click the **Edit Filters** button at the bottom of the simulation panel to display the available visible events. Toggle the **Show All/None** check box and notice how the check boxes switch from unchecked to checked or checked to unchecked, depending on the current state.

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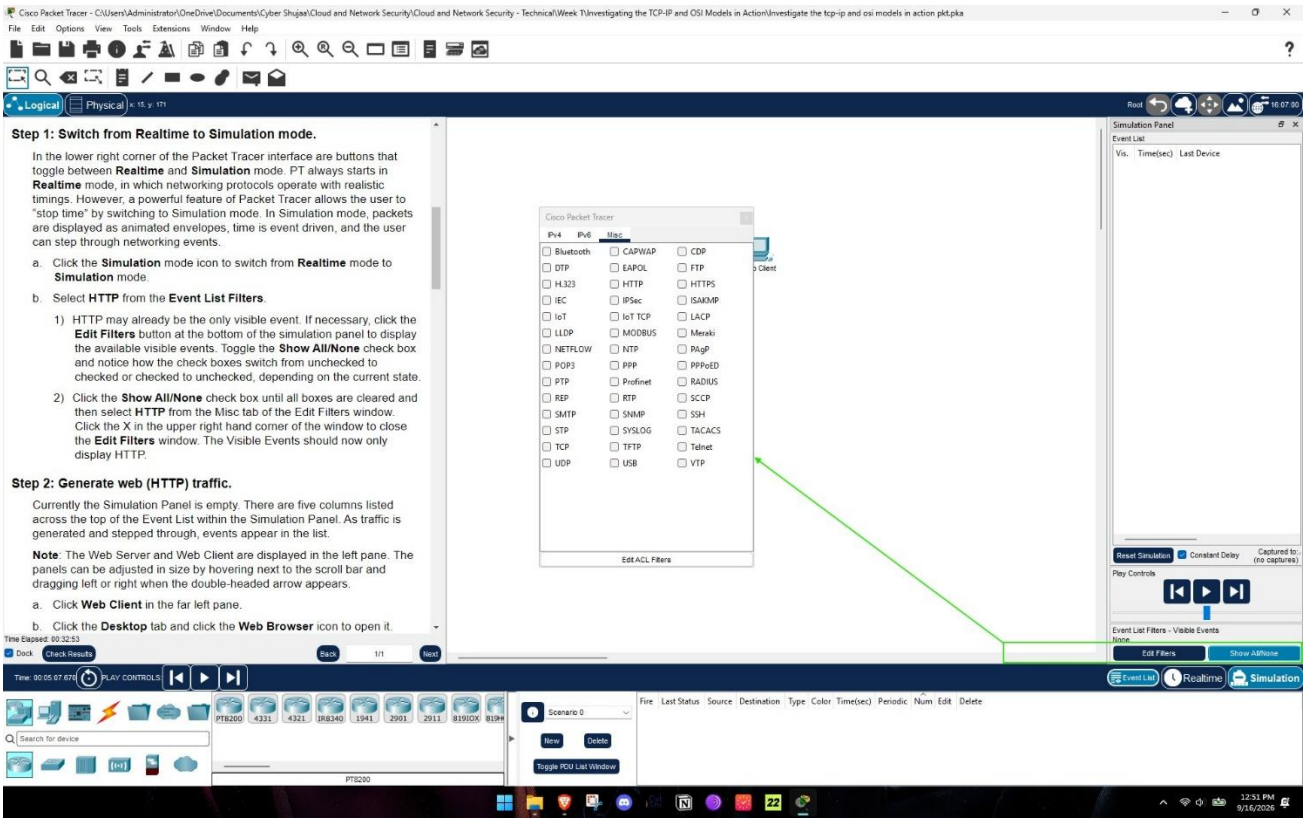


Image showing the deselection of all options after clicking Show All/ None

2) Click the **Show All/None** check box until all boxes are cleared and then select **HTTP** from the Misc tab of the Edit Filters window. Click the X in the upper right-hand corner of the window to close the **Edit Filters** window. The Visible Events should now only display HTTP.

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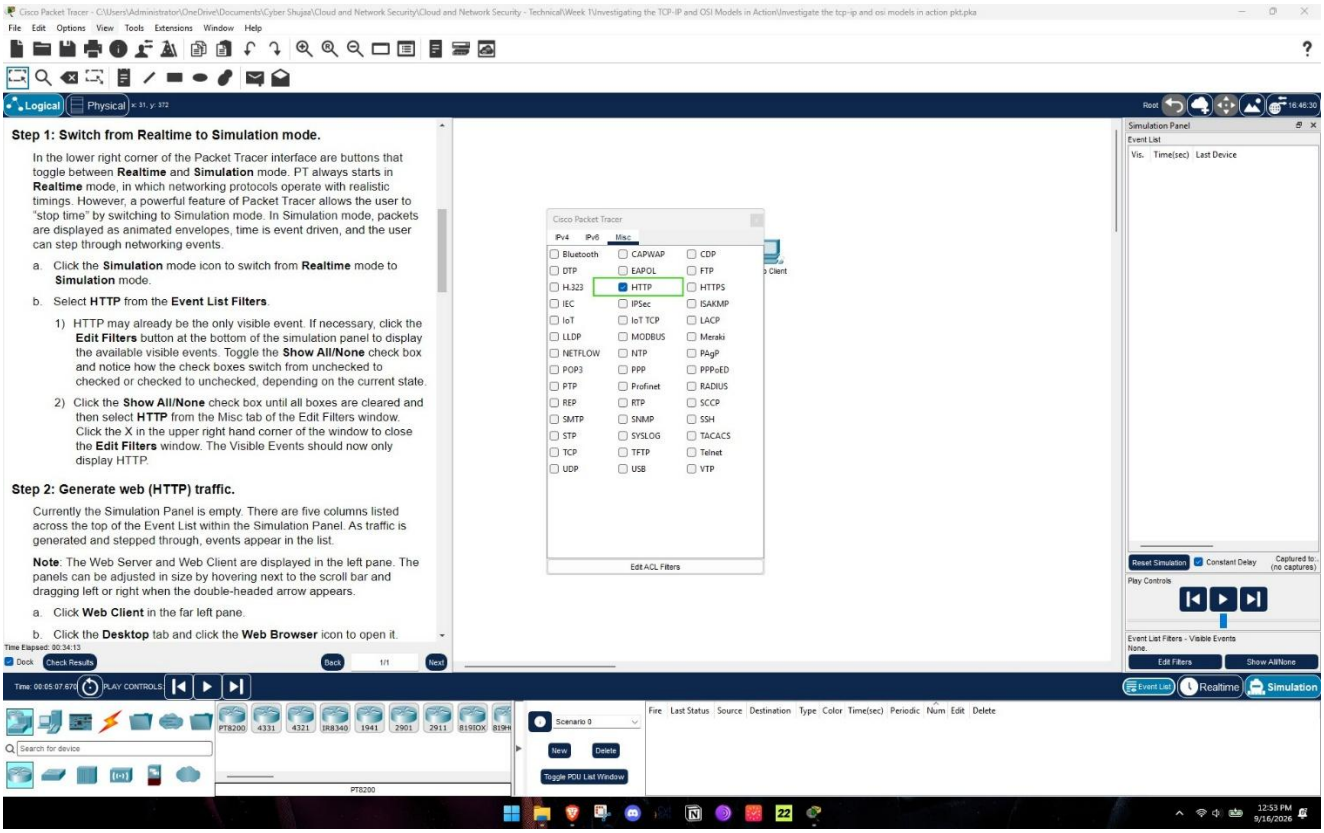


Image showing selection of the HTTP from the Misc tab

Step 2: Generate web (HTTP) traffic.

Currently the Simulation Panel is empty. There are five columns listed across the top of the Event List within the Simulation Panel. As traffic is generated and stepped through, events appear in the list.

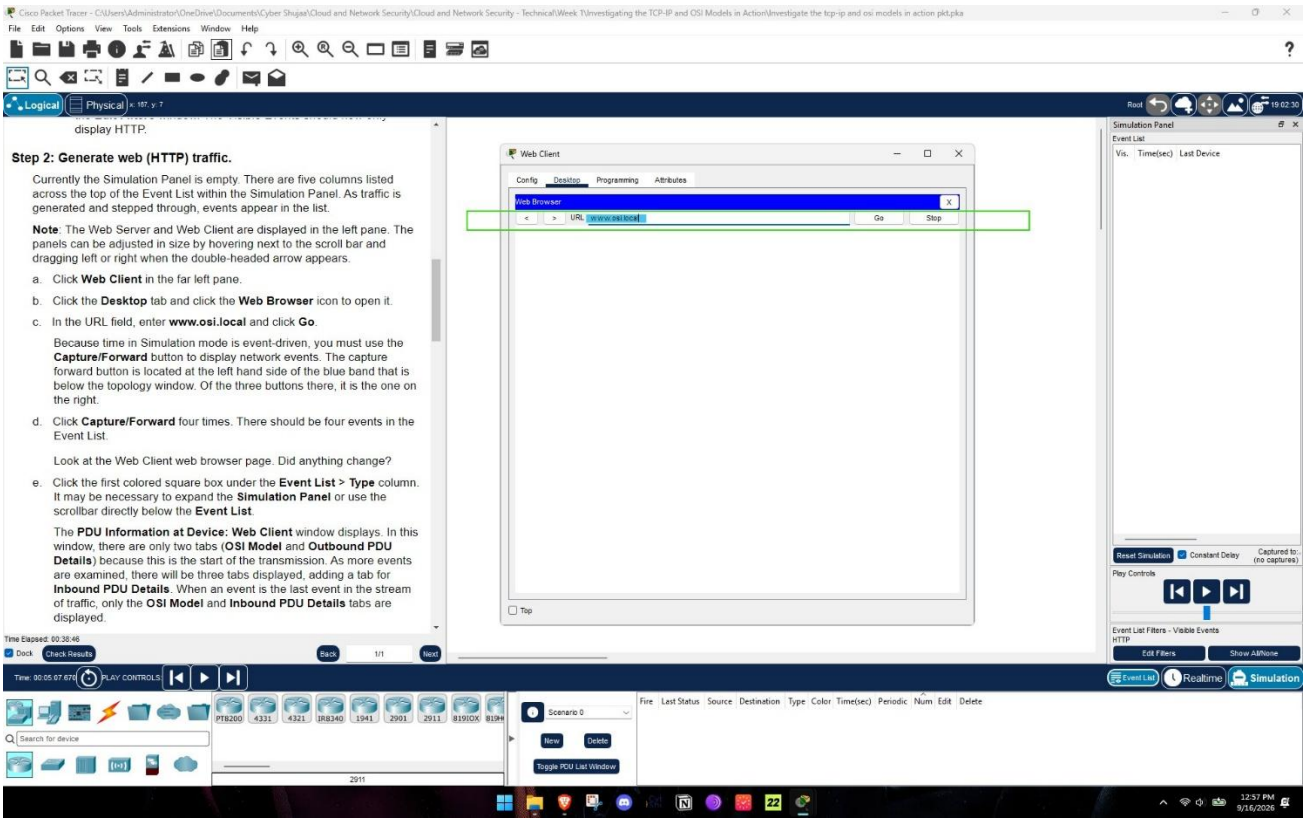
Note: The Web Server and Web Client are displayed in the left pane. The panels can be adjusted in size by hovering next to the scroll bar and dragging left or right when the double-headed arrow appears.

a. Click **Web Client** in the far-left pane.

b. Click the Desktop tab and click the Web Browser icon to open it.

c. In the URL field, enter www.osi.local and click Go.

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Because time in Simulation mode is event-driven, you must use the **Capture/Forward** button to display network events. The capture forward button is located at the left-hand side of the blue band that is below the topology window. Of the three buttons there, it is the one on the right.

d. Click **Capture/Forward** four times. There should be four events in the Event List.

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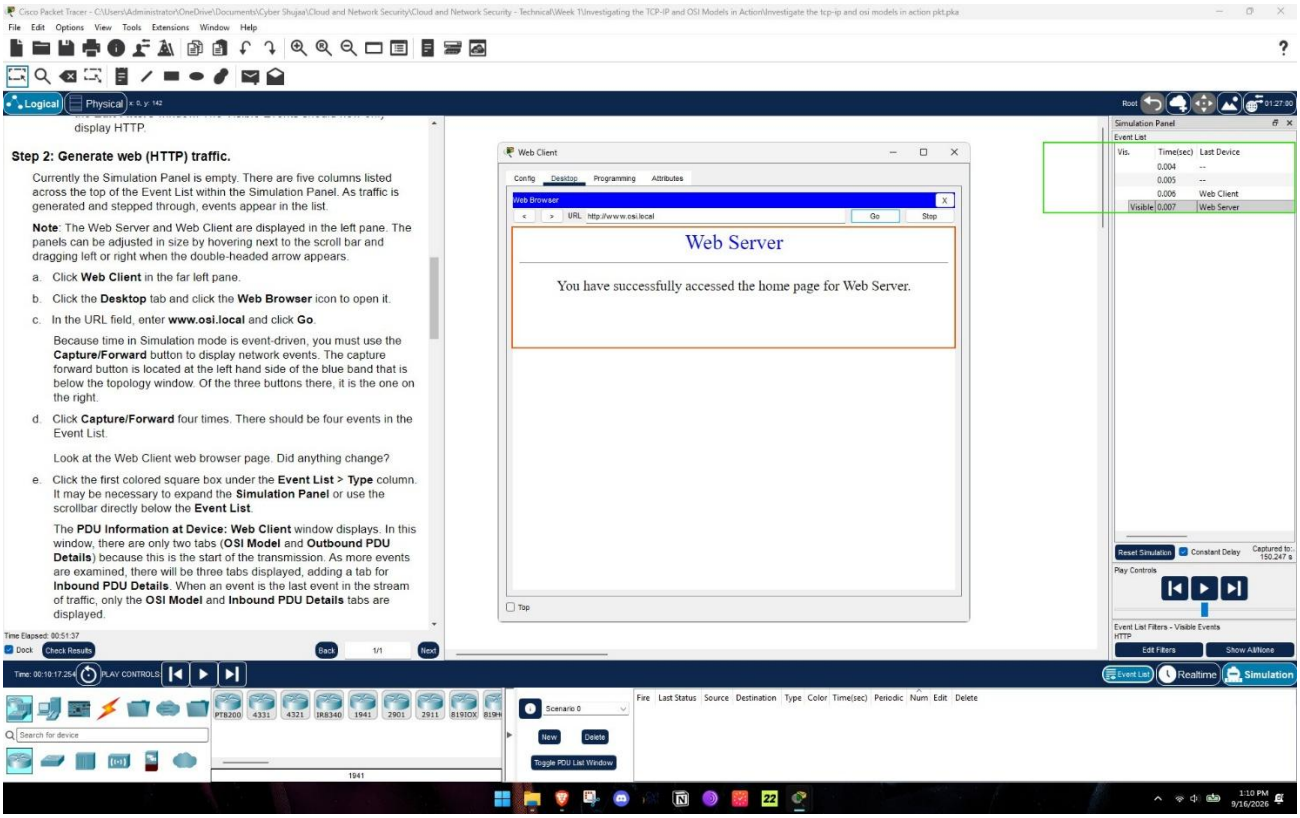


Image showing the four events in the events list, in the green box at the top right side.

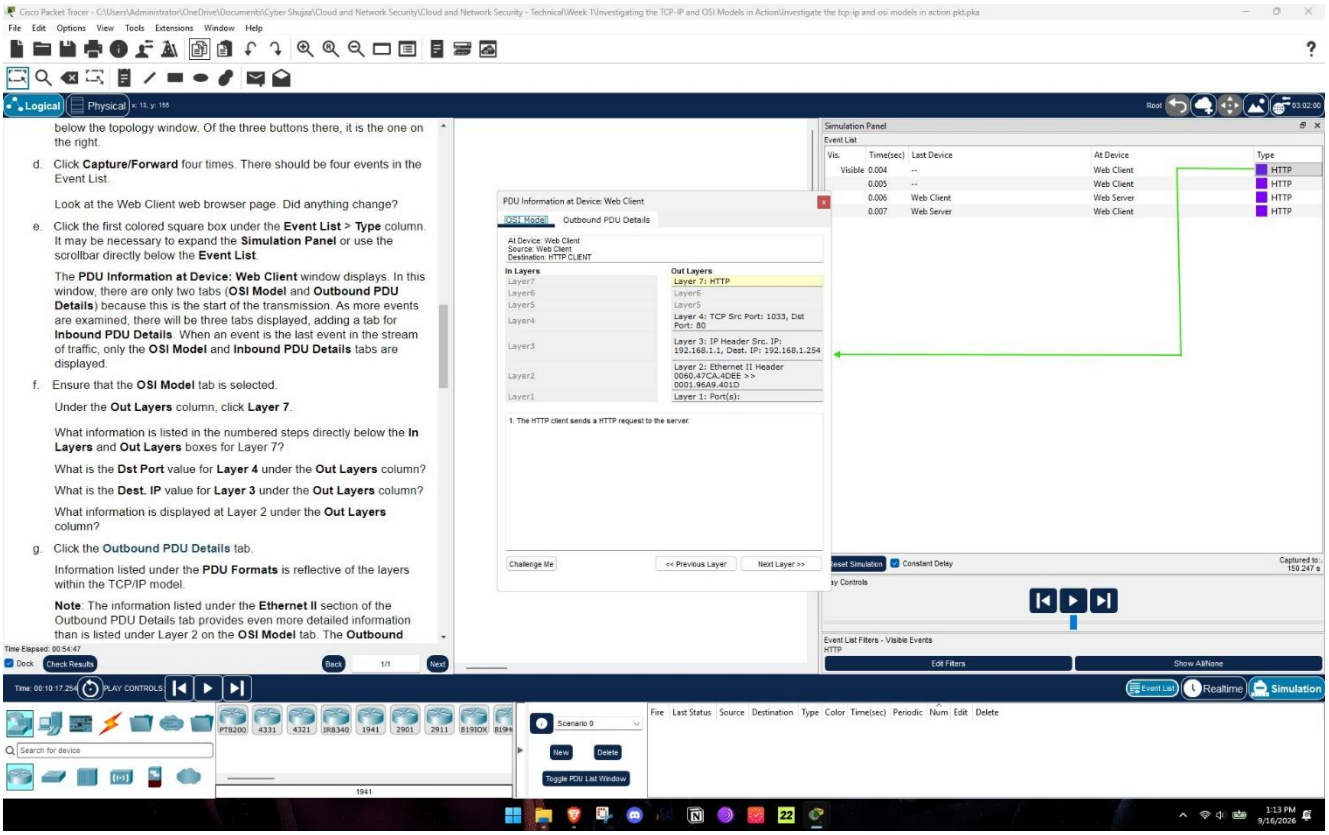
Look at the Web Client web browser page. Did anything change?

Yes, the web page was returned from the web server.

Step 3: Explore the contents of the HTTP packet.

a. Click the first coloured square box under the **Event List** > **Type** column. It may be necessary to expand the **Simulation Panel** or use the scrollbar directly below the **Event List**.

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*The **PDU Information at Device: Web Client** window displays. In this window, there are only two tabs (**OSI Model** and **Outbound PDU Details**) because this is the start of the transmission. As more events are examined, there will be three tabs displayed, adding a tab for **Inbound PDU Details**. When an event is the last event in the stream of traffic, only the **OSI Model** and **Inbound PDU Details** tabs are displayed.*

b. Ensure that the **OSI Model** tab is selected.

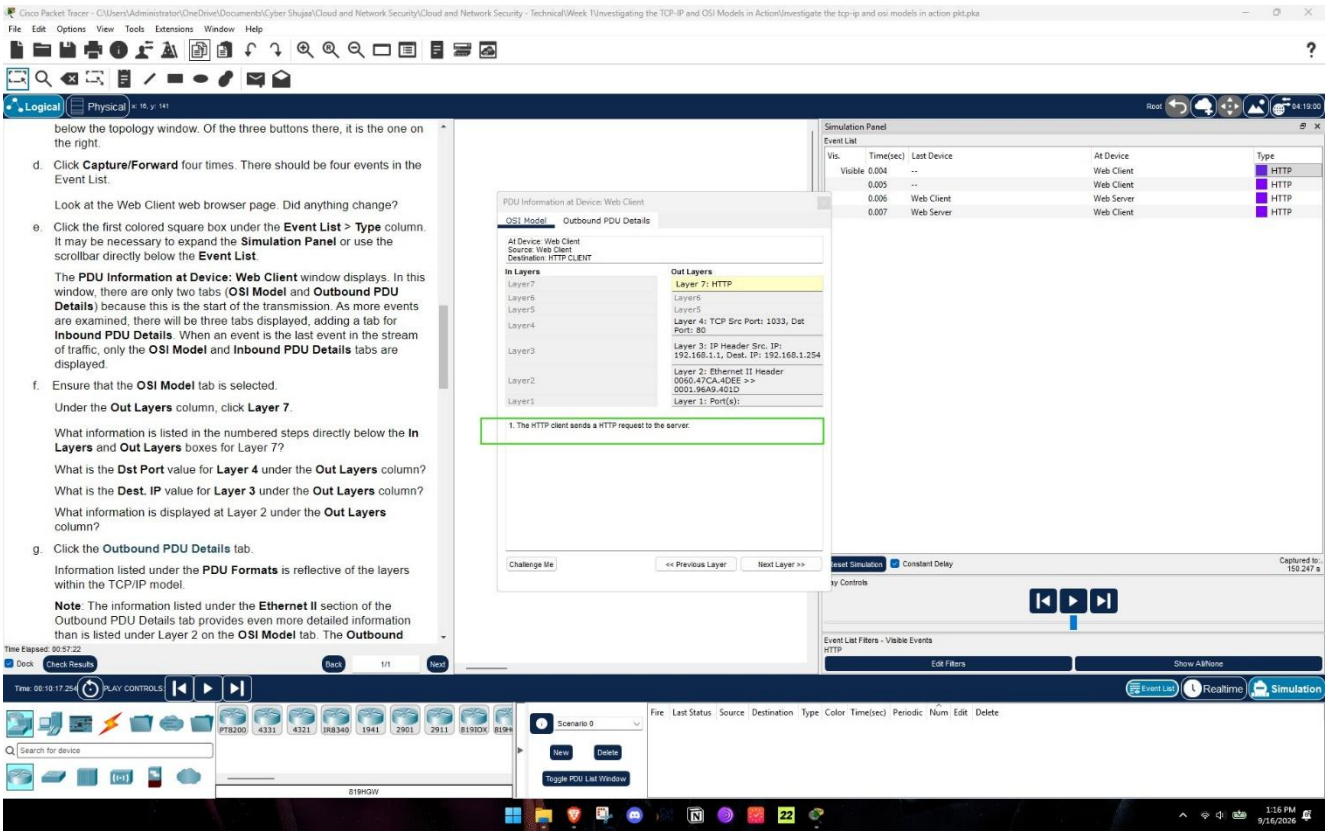
Under the **Out Layers** column, click **Layer 7**.

Questions

What information is listed in the numbered steps directly below the In Layers and Out Layers boxes for Layer 7?

"The HTTP client sends a HTTP request to the server." This is shown in the image below:

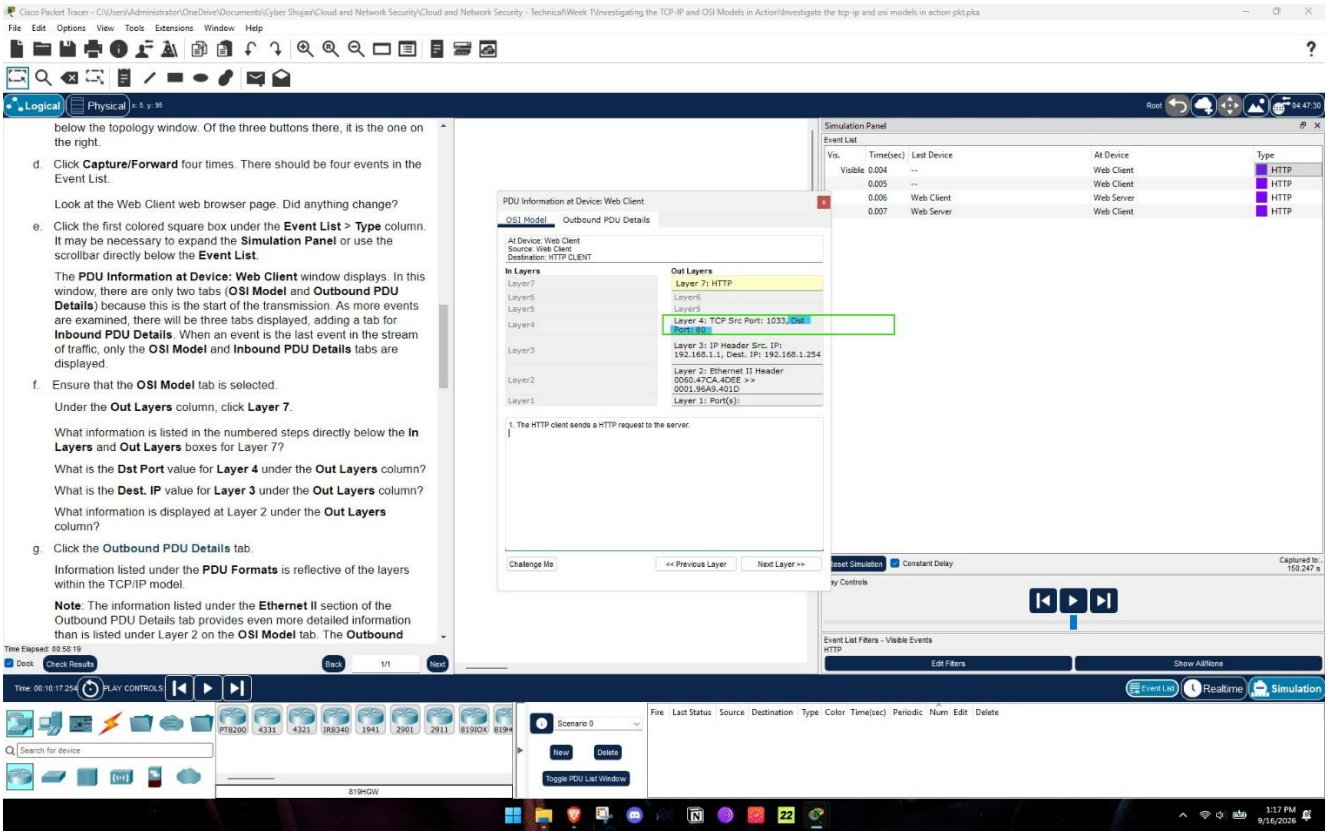
Lab: Investigating TCP/IP and OSI Models in Action



What is the Dst Port value for Layer 4 under the Out Layers column?

The DST port value is 80, shown in the image below:

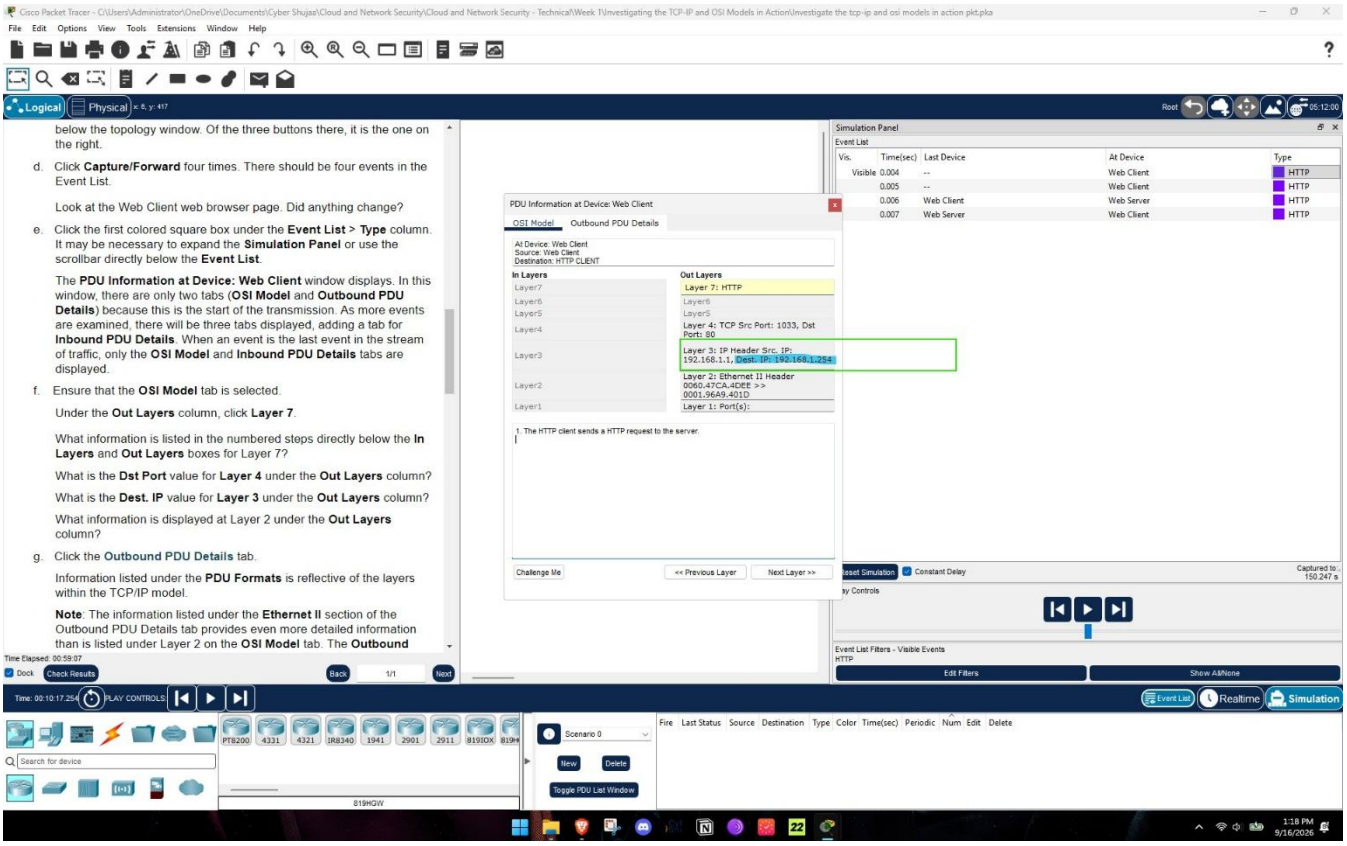
Lab: Investigating TCP/IP and OSI Models in Action



What is the Dest. IP value for Layer 3 under the Out Layers column?

The Dest. IP is 192.168.1.254, as evidenced in the image below:

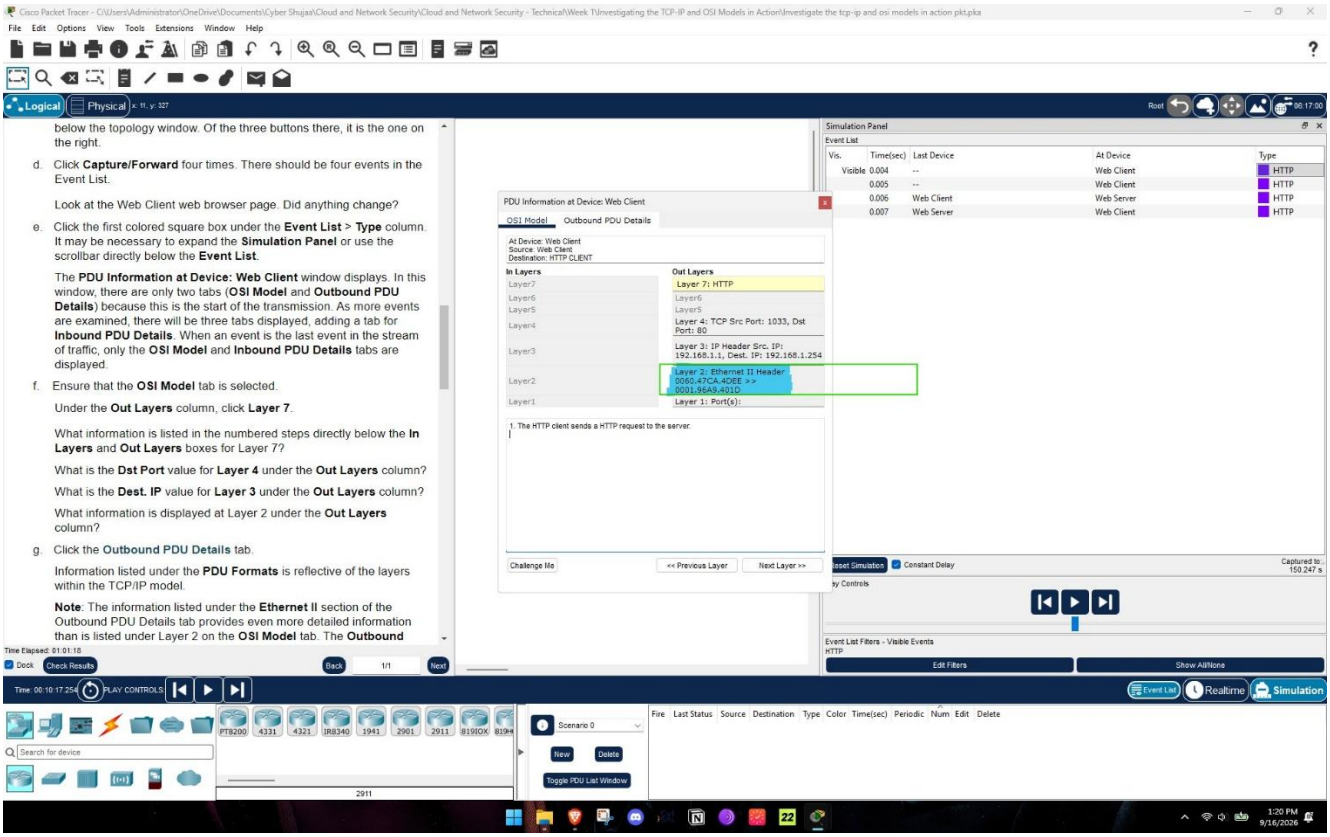
Lab: Investigating TCP/IP and OSI Models in Action



What information is displayed at Layer 2 under the Out Layers column?

Layer 2 Ethernet II Header and inbound and outbound MAC addresses. This is shown below:

Lab: Investigating TCP/IP and OSI Models in Action



c. Click the **Outbound PDU Details** tab.

Information listed under the **PDU Formats** is reflective of the layers within the TCP/IP model.

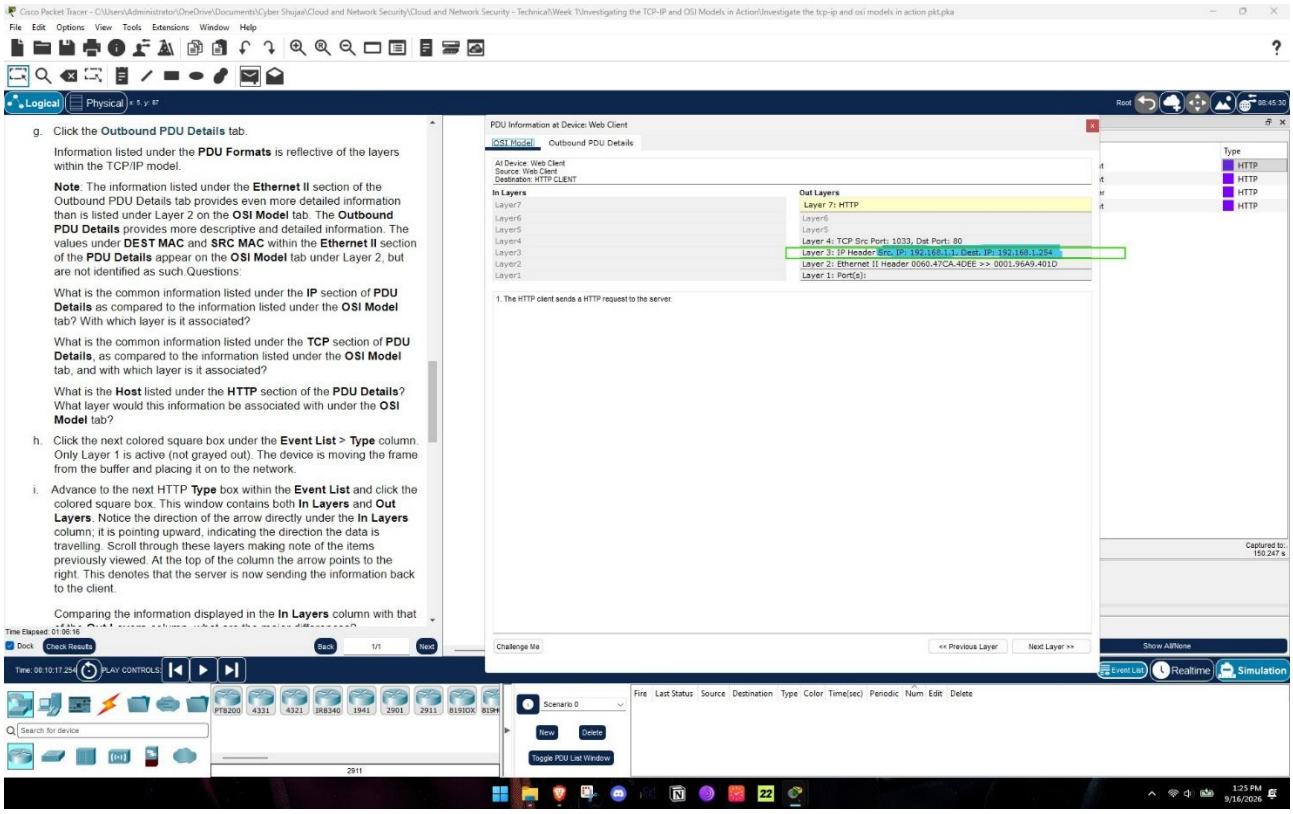
Note: The information listed under the **Ethernet II** section of the Outbound PDU Details tab provides even more detailed information than is listed under Layer 2 on the **OSI Model** tab. The **Outbound PDU Details** provides more descriptive and detailed information. The values under **DEST MAC** and **SRC MAC** within the **Ethernet II** section of the **PDU Details** appear on the **OSI Model** tab under Layer 2, but are not identified as such.

Questions

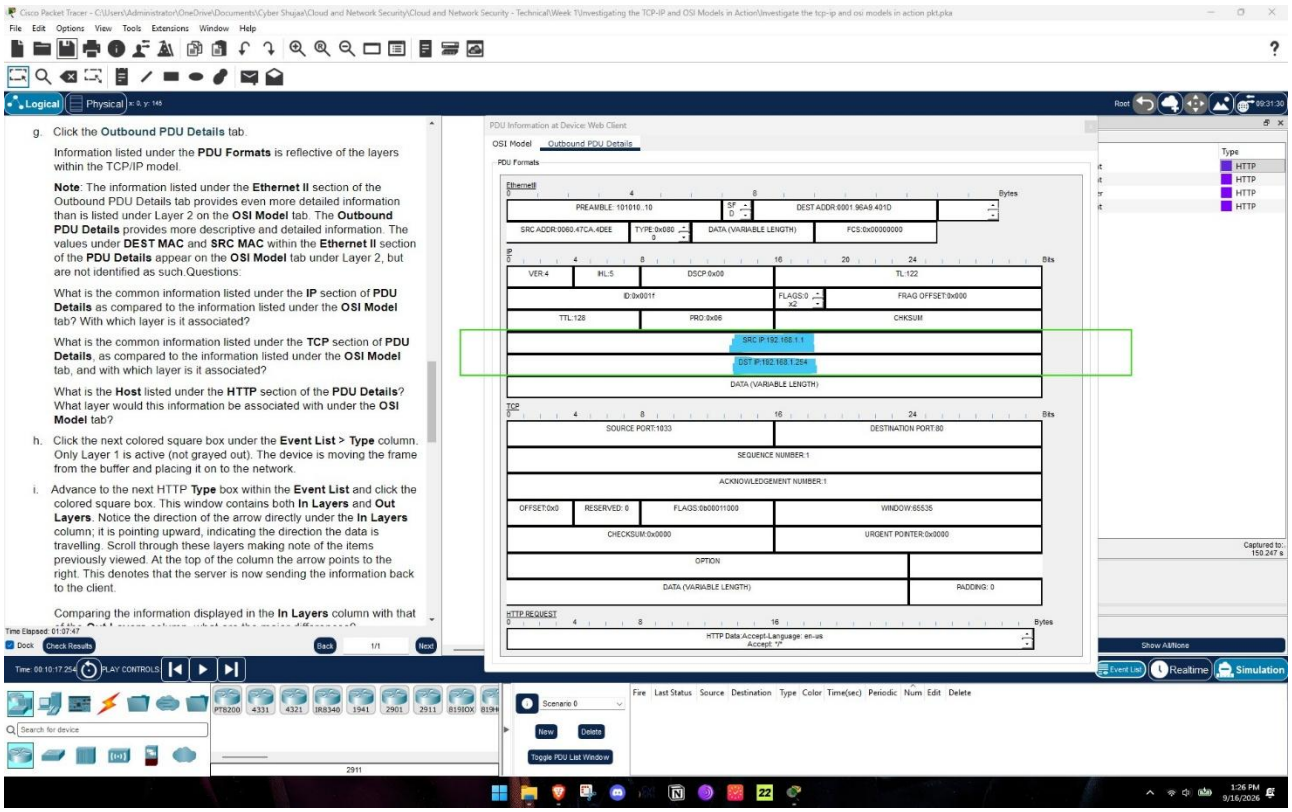
What is the common information listed under the IP section of PDU Details as compared to the information listed under the OSI Model tab? With which layer is it associated?

The source IP and the destination IP at layer 3, as shown below:

Lab: Investigating TCP/IP and OSI Models in Action



First image showing the source and destination Ips at layer 3

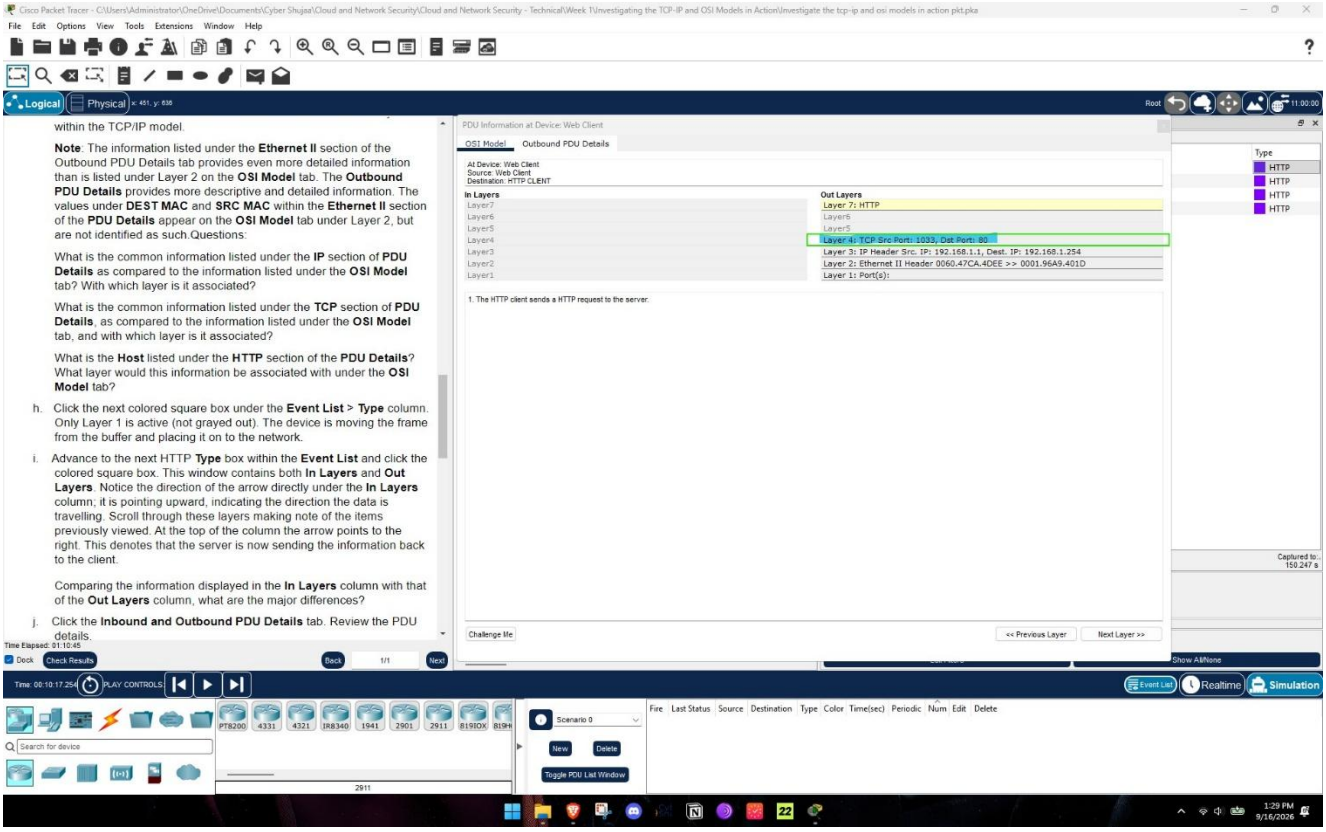


Lab: Investigating TCP/IP and OSI Models in Action

The same source and destination IPs in this image.

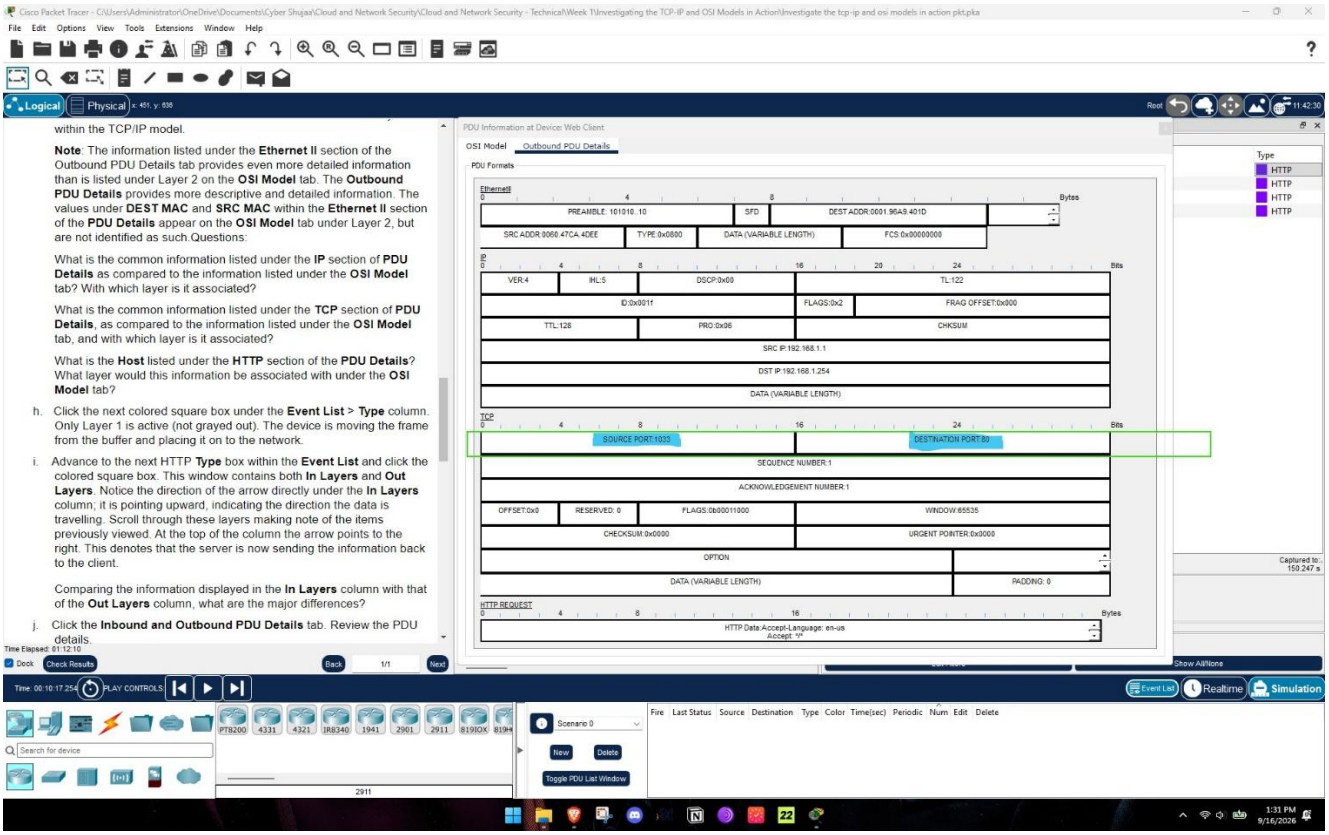
What is the common information listed under the TCP section of PDU Details, as compared to the information listed under the OSI Model tab, and with which layer is it associated?

The common information is the source and destination ports, and it is associated with Layer 4. This is shown in the two images below:



First image showing the ports in the OSI model tab

Lab: Investigating TCP/IP and OSI Models in Action



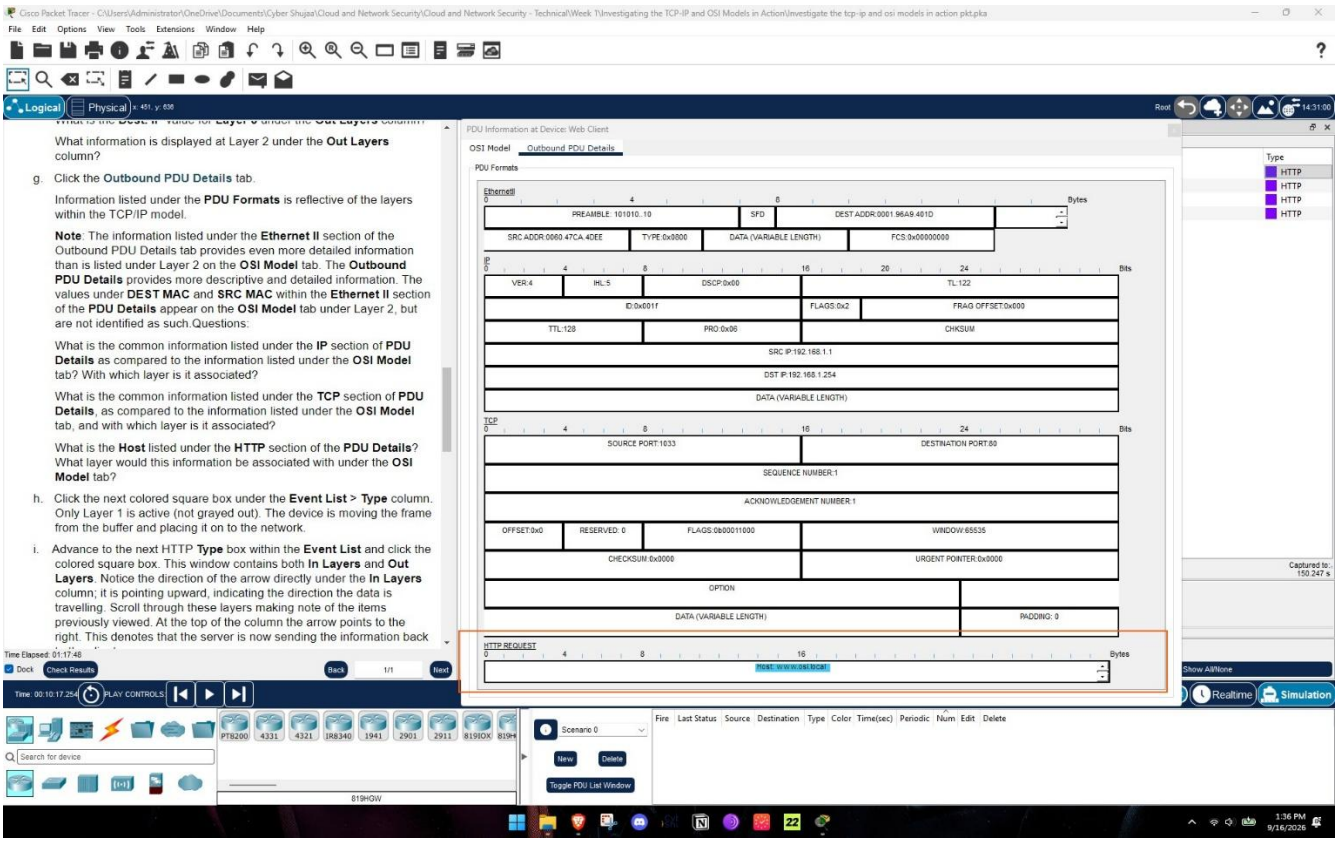
Second image sowing the same ports in the Outlined PDU Details tab.

What is the Host listed under the HTTP section of the PDU Details? What layer would this information be associated with under the OSI Model tab?

www.osi.local

The layer that would be associated with is Layer 7.

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d. Click the next coloured square box under the **Event List > Type** column. Only Layer 1 is active (not greyed out). The device is moving the frame from the buffer and placing it on to the network.

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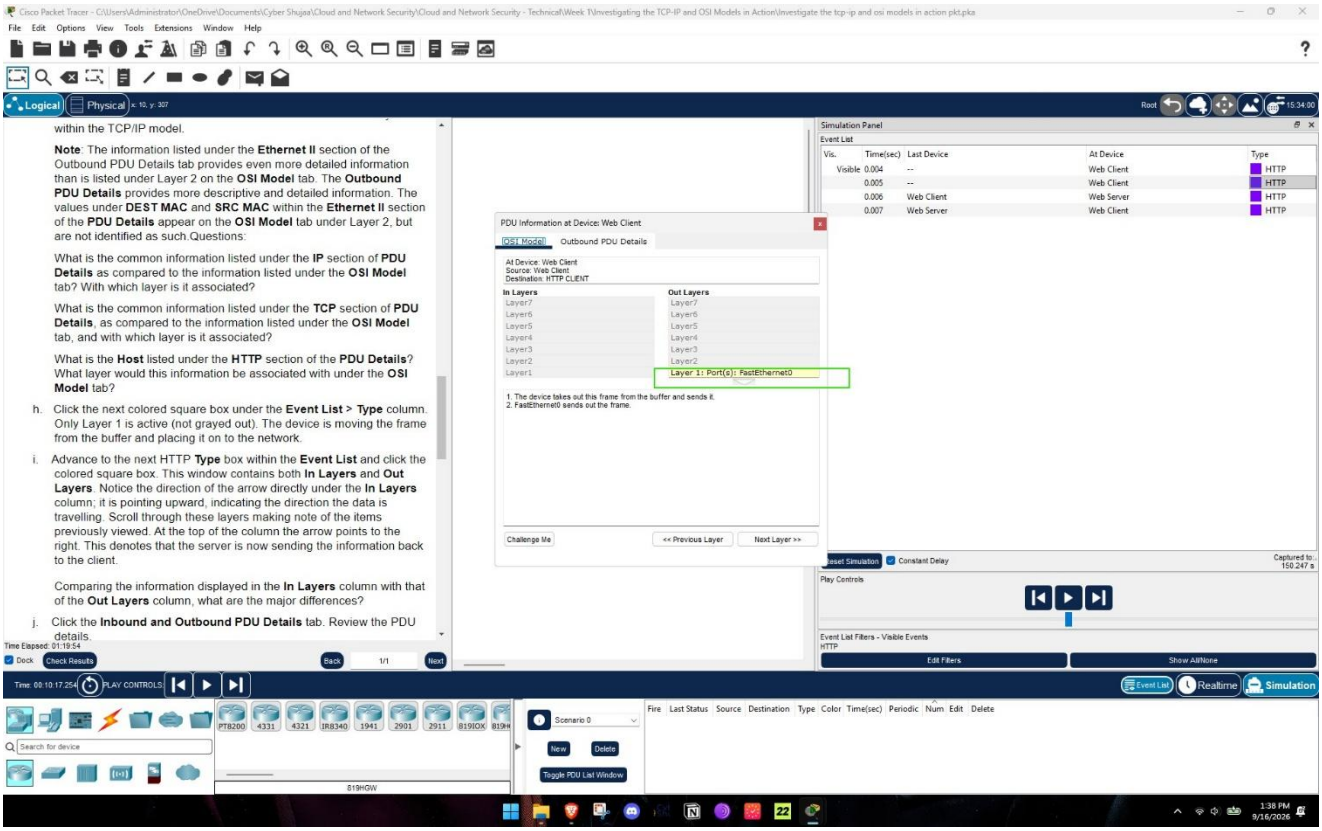


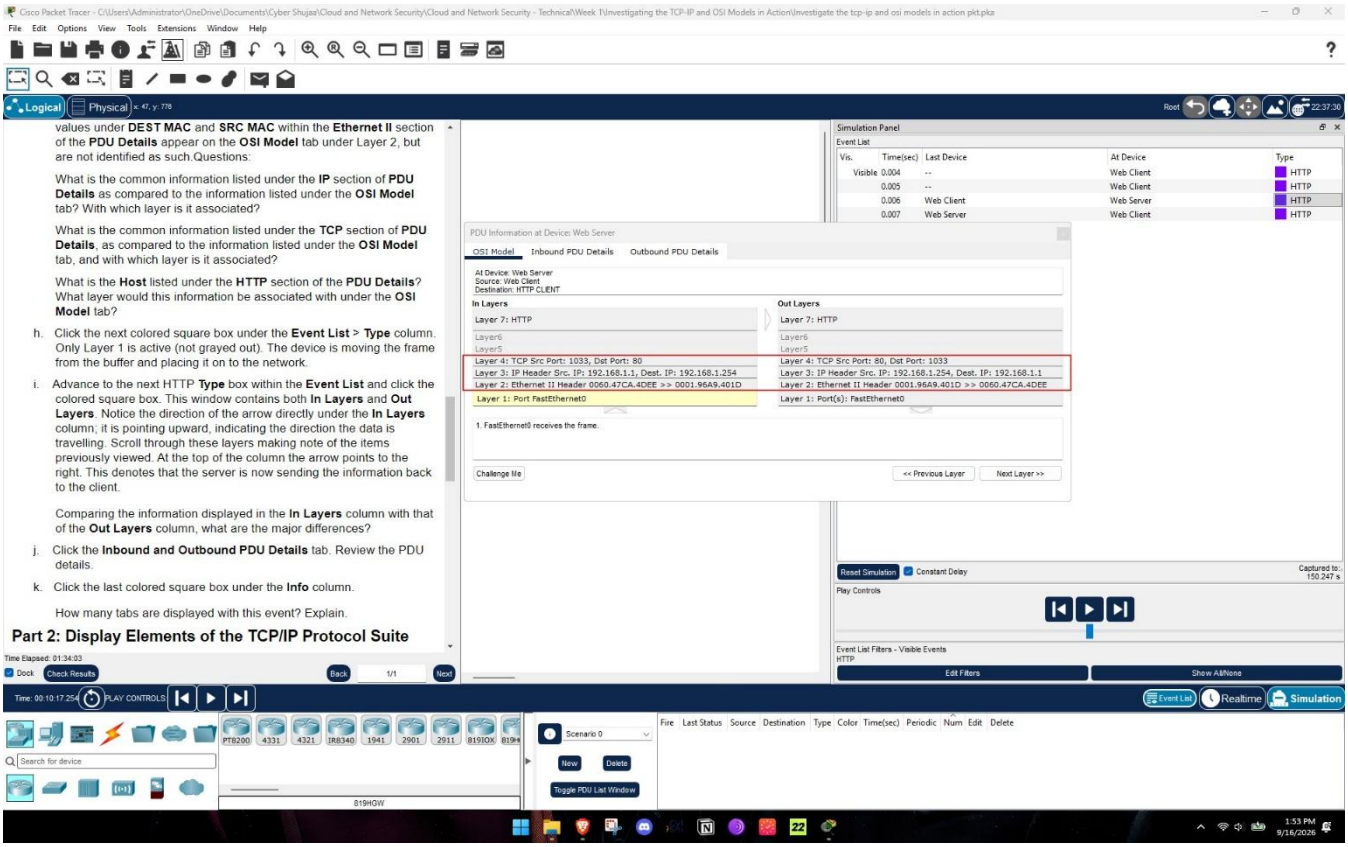
Image showing that only Layer 1 is active

e. Advance to the next HTTP **Type** box within the **Event List** and click the coloured square box. This window contains both **In Layers** and **Out Layers**. Notice the direction of the arrow directly under the **In Layers** column; it is pointing upward, indicating the direction the data is travelling. Scroll through these layers making note of the items previously viewed. At the top of the column the arrow points to the right. This denotes that the server is now sending the information back to the client.

Comparing the information displayed in the In Layers column with that of the Out Layers column, what are the major differences?

The source and destination ports, Ips and MAC addresses have been swapped.

Lab: Investigating TCP/IP and OSI Models in Action



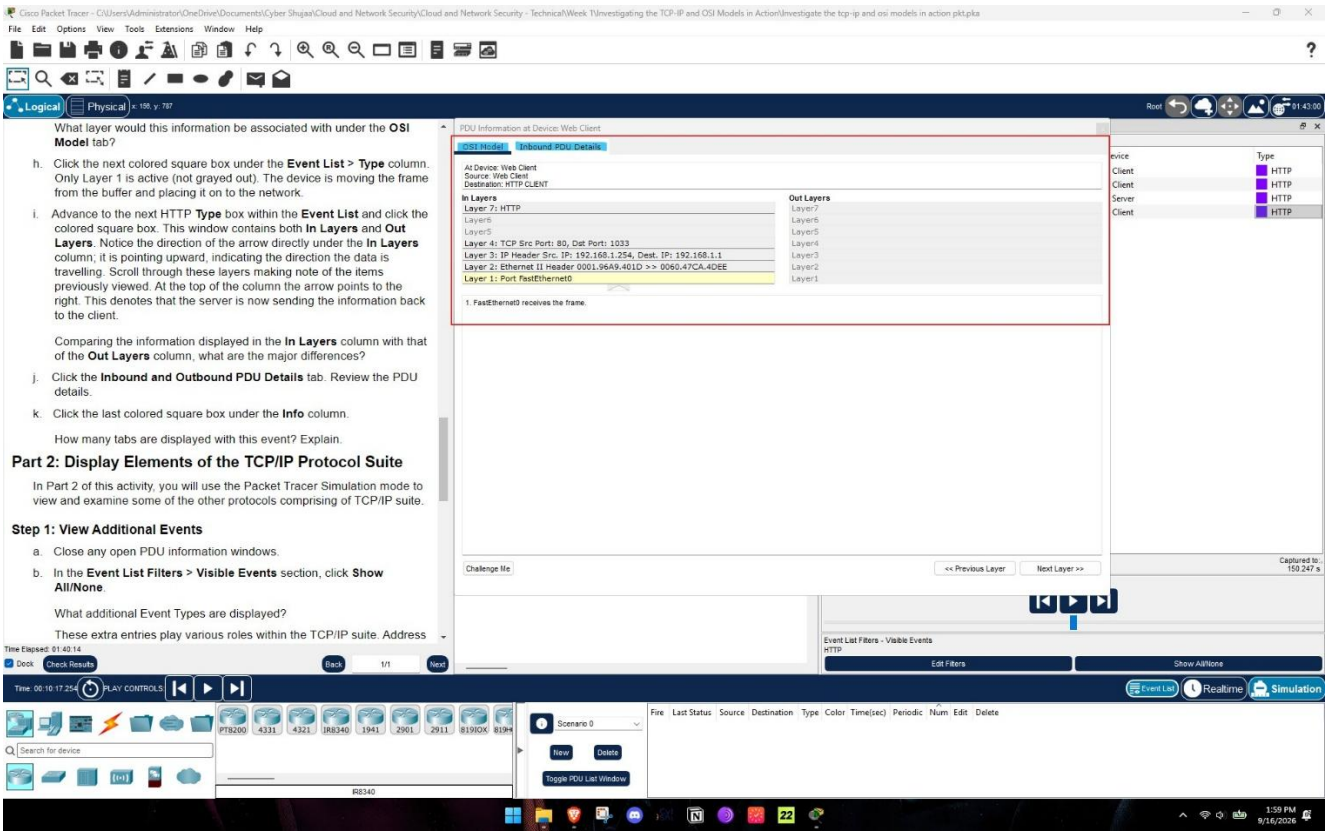
f. Click the **Inbound and Outbound PDU Details** tab. Review the PDU details.

g. Click the last-coloured square box under the **Info** column.

How many tabs are displayed with this event? Explain.

There are 2 tabs displayed: one for the OSI model and one for the Inbound PDU Details. This is because it is the receiving device.

Lab: Investigating TCP/IP and OSI Models in Action



Part 2: Display Elements of the TCP/IP Protocol Suite

Step 1: View Additional Events

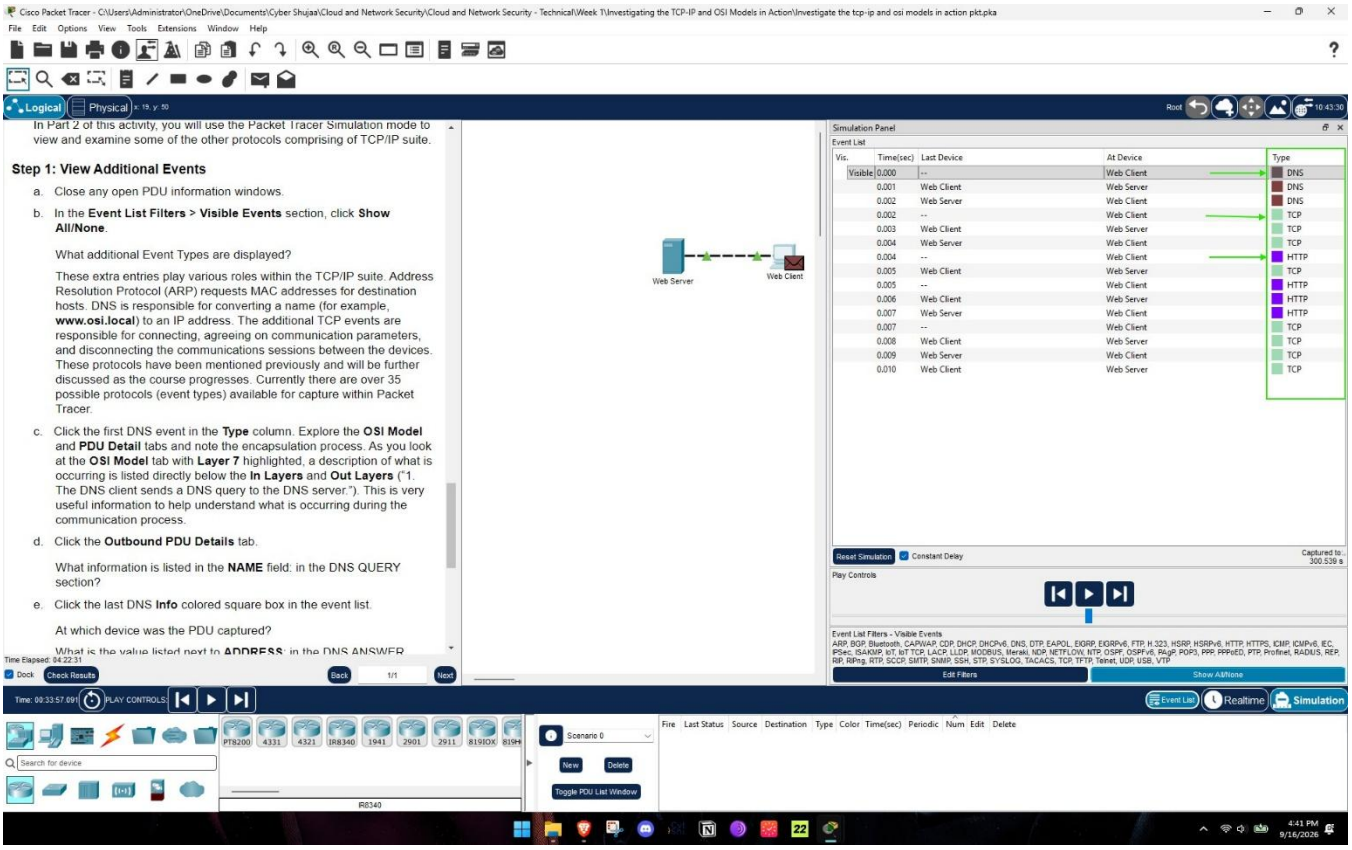
a. Close any open PDU information windows.

b. In the Event List Filters > Visible Events section, click Show All/None.

What additional Event Types are displayed?

Having looked at HTTP previously, the additional event types displayed are DNS and TCP.

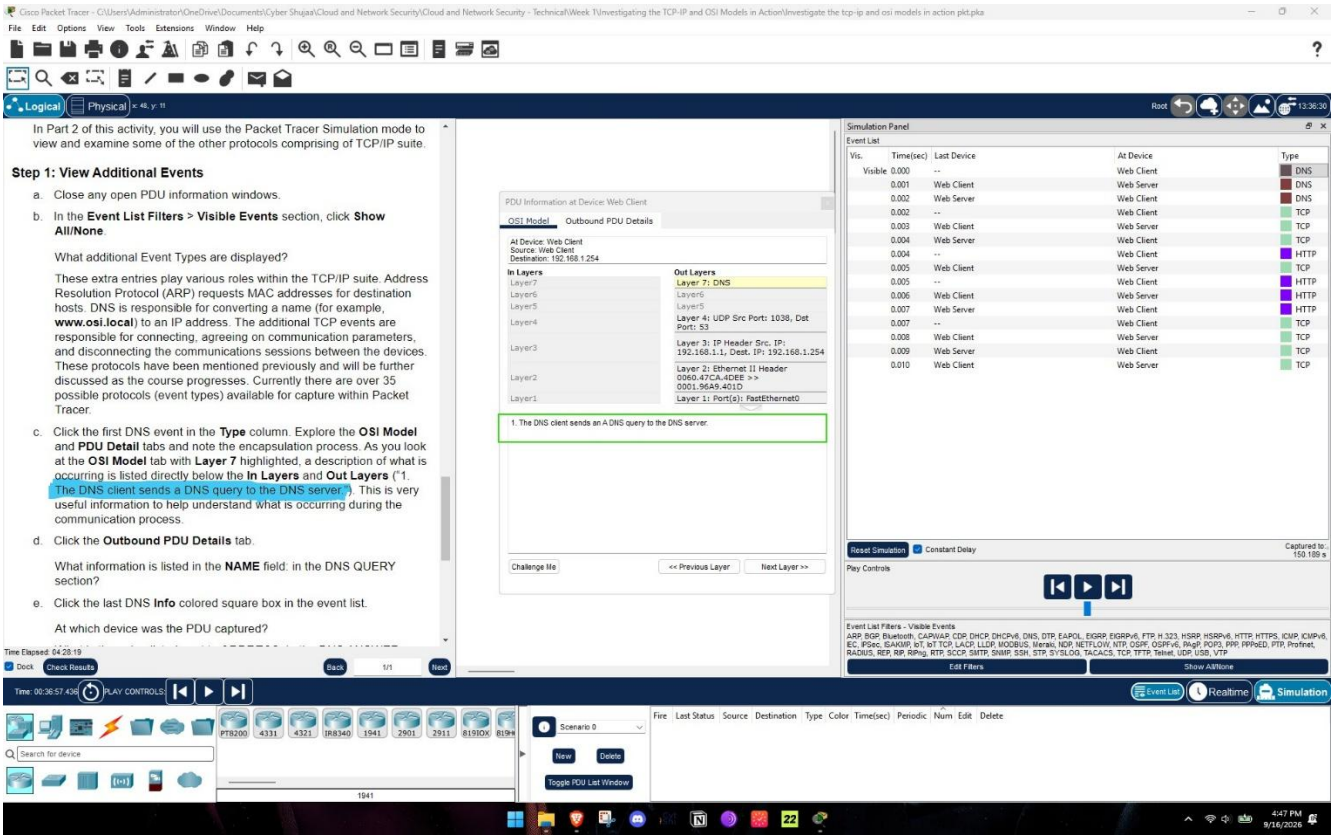
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*These extra entries play various roles within the TCP/IP suite. Address Resolution Protocol (ARP) requests MAC addresses for destination hosts. DNS is responsible for converting a name (for example, **www.osi.local**) to an IP address. The additional TCP events are responsible for connecting, agreeing on communication parameters, and disconnecting the communications sessions between the devices. These protocols have been mentioned previously and will be further discussed as the course progresses. Currently there are over 35 possible protocols (event types) available for capture within Packet Tracer.*

c. Click the first DNS event in the **Type** column. Explore the **OSI Model** and **PDU Detail** tabs and note the encapsulation process. As you look at the **OSI Model** tab with **Layer 7** highlighted, a description of what is occurring is listed directly below the **In Layers** and **Out Layers** ("1. The DNS client sends a DNS query to the DNS server."). This is very useful information to help understand what is occurring during the communication process.

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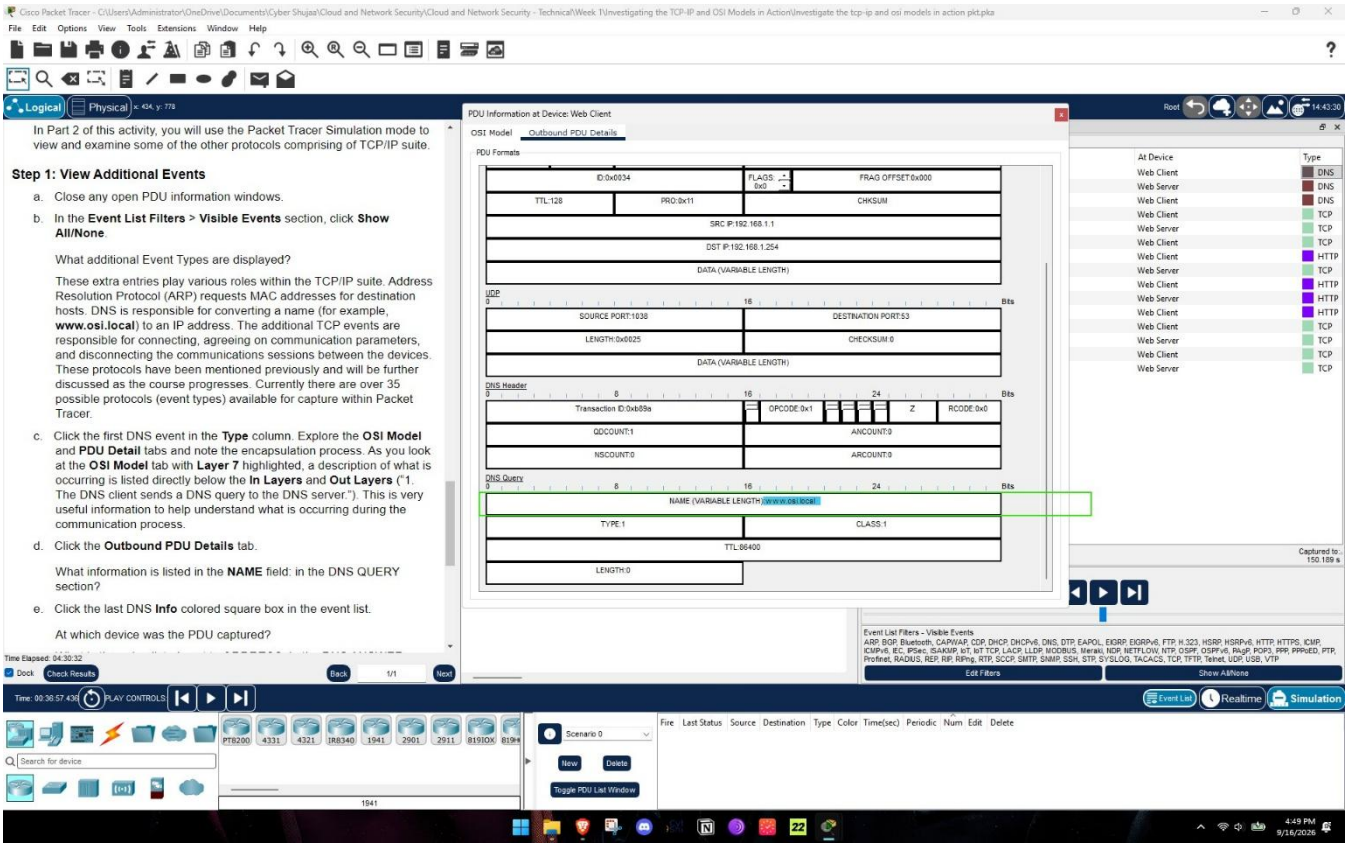


d. Click the **Outbound PDU Details** tab.

What information is listed in the NAME field: in the DNS QUERY section?

www.osi.local. This is shown below:

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e. Click the last DNS Info coloured square box in the event list.

At which device was the PDU captured?

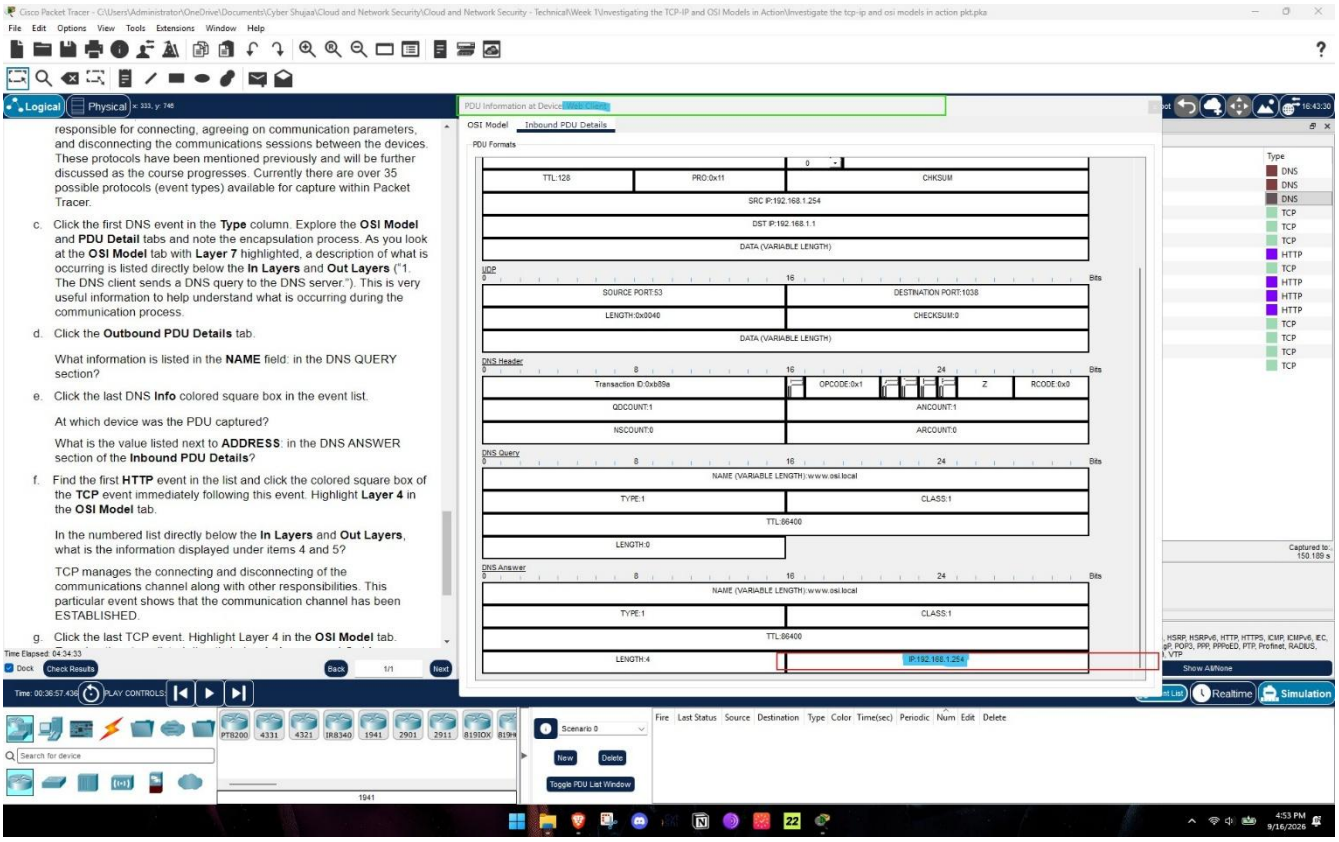
The Web Client's device

What is the value listed next to ADDRESS: in the DNS ANSWER section of the Inbound PDU Details?

192.168.1.254.

The answers to both questions are outlined in the image below (highlighted):

Lab: Investigating TCP/IP and OSI Models in Action



f. Find the first HTTP event in the list and click the coloured square box of the TCP event immediately following this event. Highlight Layer 4 in the OSI Model tab.

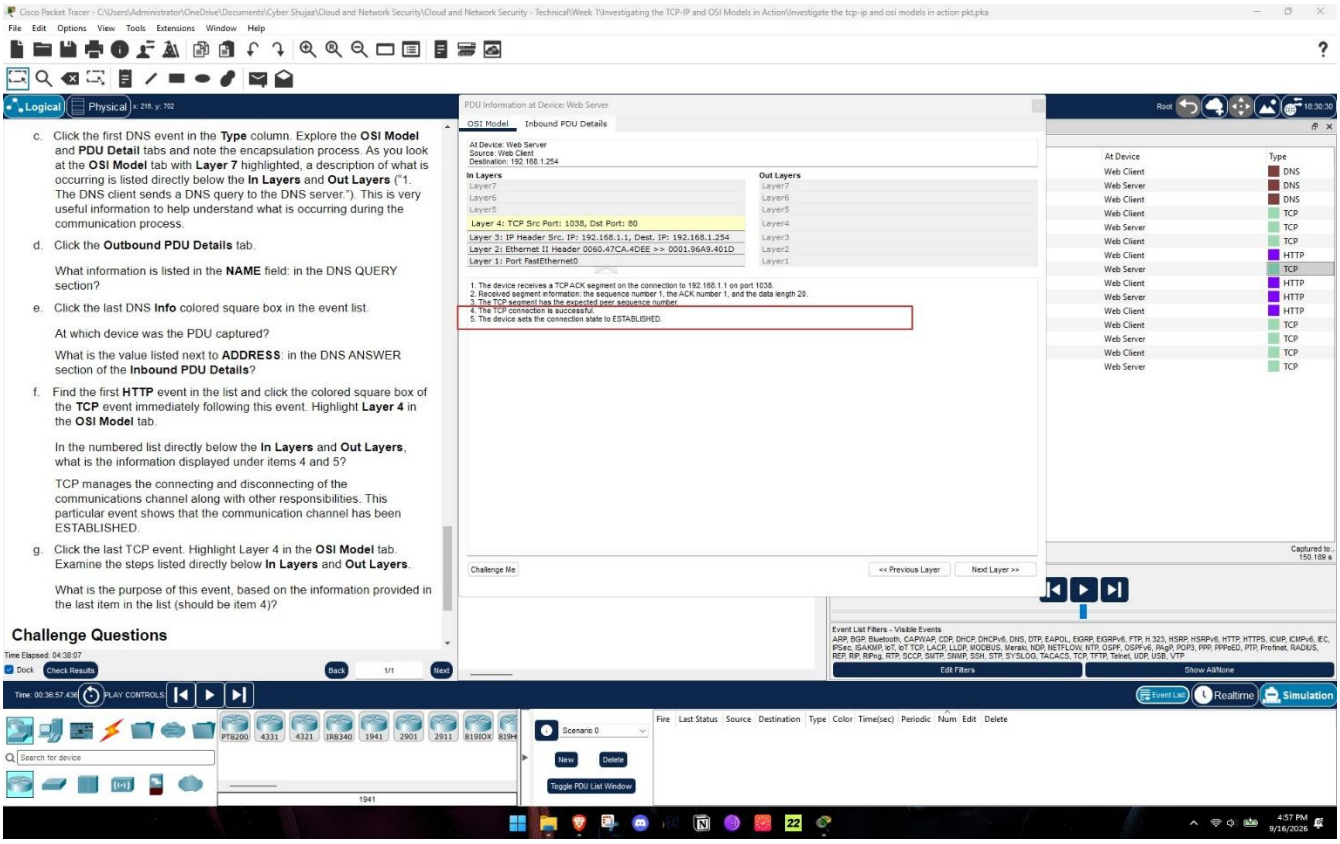
In the numbered list directly below the In Layers and Out Layers, what is the information displayed under items 4 and 5?

"4. The TCP connection is successful."

"5. The device sets the connection state to ESTABLISHED."

This is shown below:

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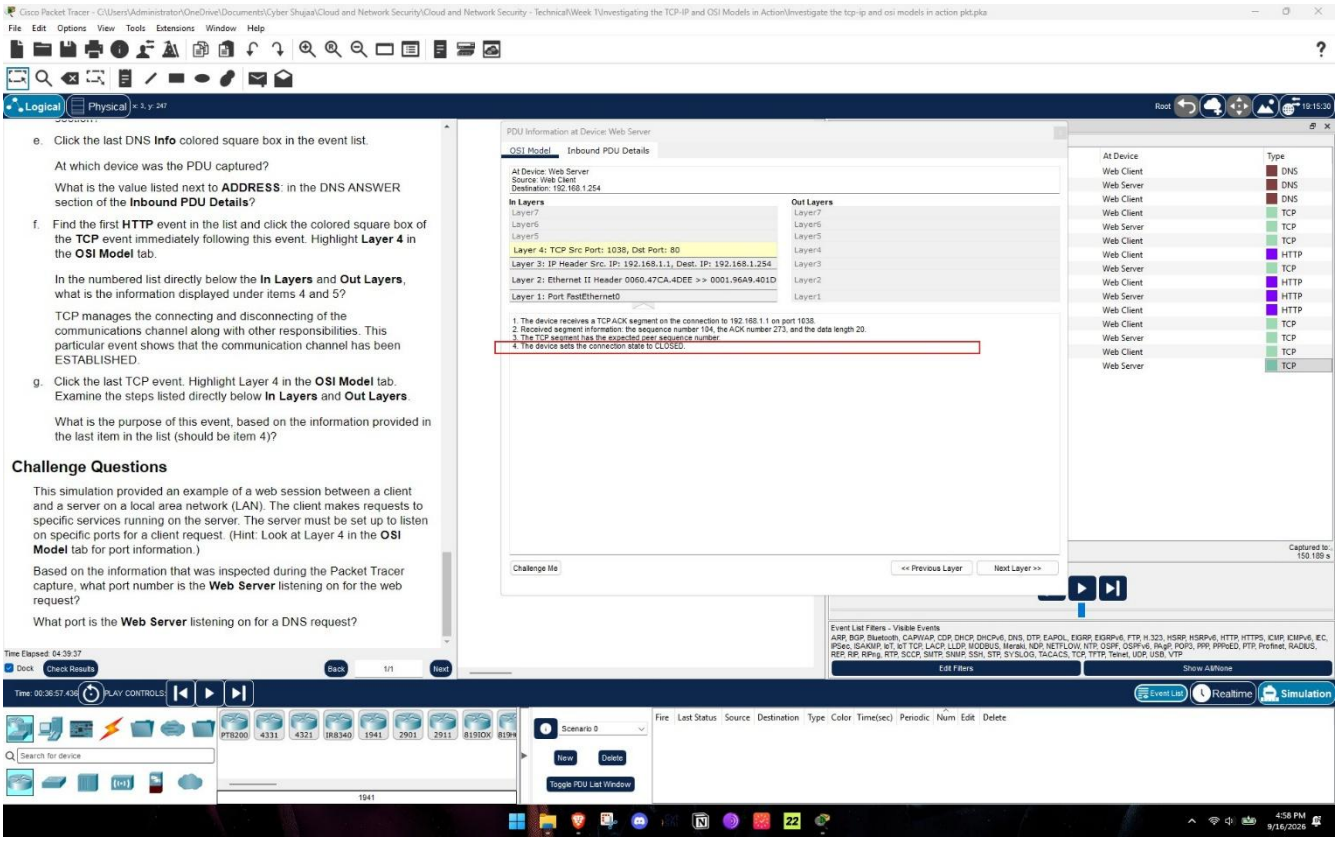
TCP manages the connecting and disconnecting of the communications channel along with other responsibilities. This particular event shows that the communication channel has been ESTABLISHED.

g. Click the last TCP event. Highlight Layer 4 in the OSI Model tab. Examine the steps listed directly below In Layers and Out Layers.

What is the purpose of this event, based on the information provided in the last item in the list (should be item 4)?

The purpose is closing the connection, as shown below:

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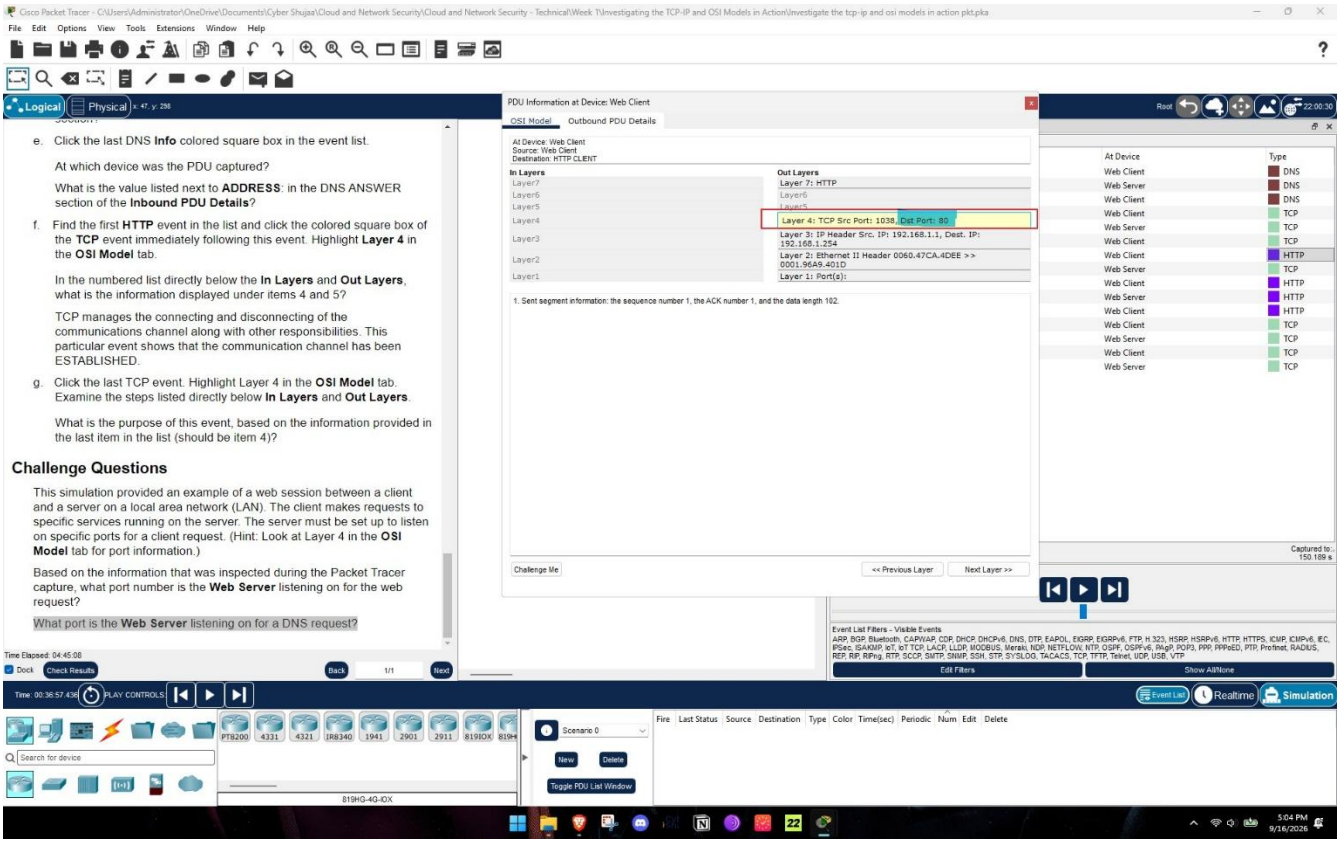
Challenge Questions

This simulation provided an example of a web session between a client and a server on a local area network (LAN). The client makes requests to specific services running on the server. The server must be set up to listen on specific ports for a client request. (Hint: Look at Layer 4 in the **OSI Model** tab for port information.)

Based on the information that was inspected during the Packet Tracer capture, what port number is the Web Server listening on for the web request?

The port number is port 80 in layer 4 DST port.

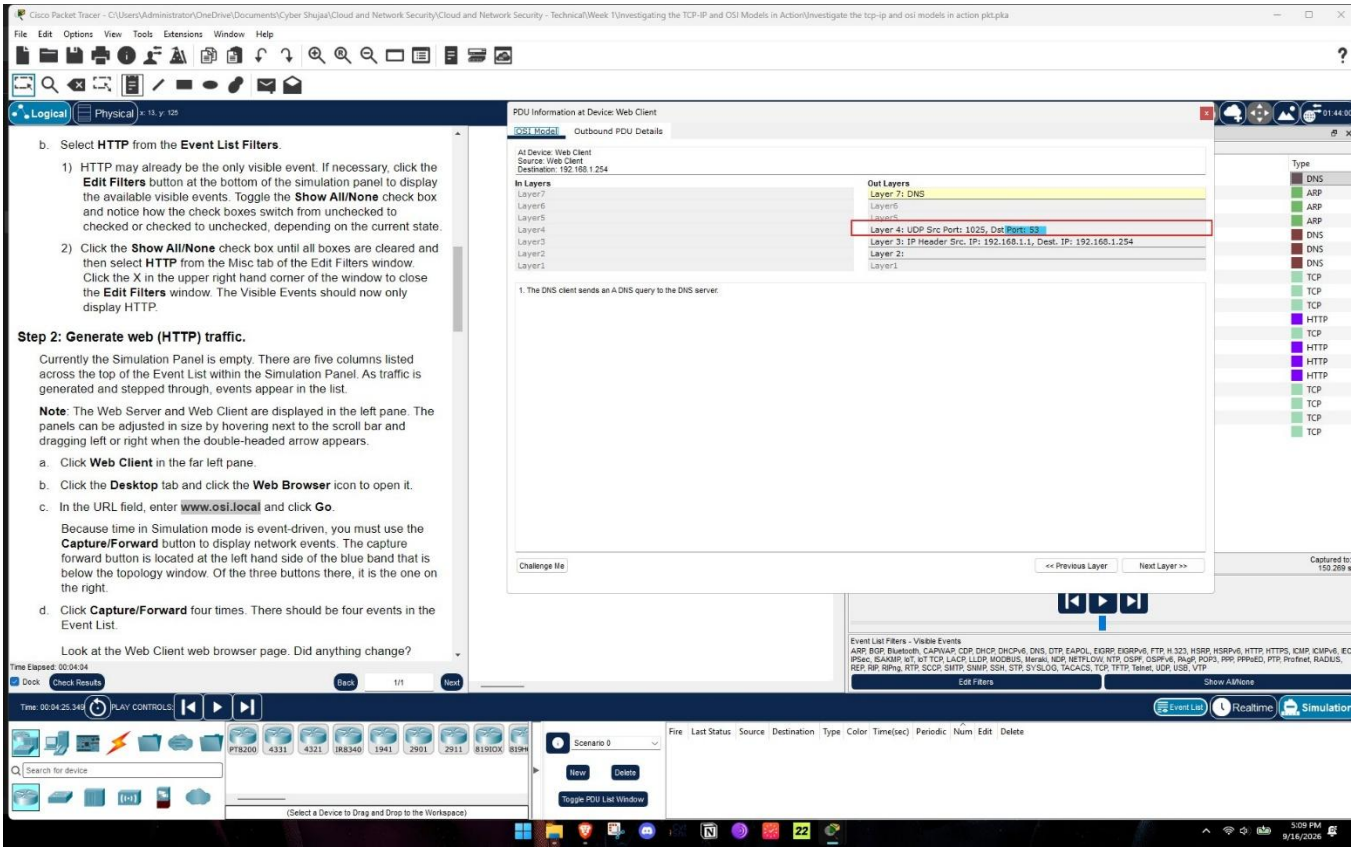
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What port is the Web Server listening on for a DNS request?

The web server is listening on for a DNS request on port 53 in layer 4 DST port

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Conclusion

The lab successfully demonstrates the practical interaction of network protocols by analyzing captured PDU details and tracking communication states between devices. Through the step-by-step examination of HTTP, DNS, and TCP events, the findings verify how data is encapsulated across layers, how source and destination parameters (MAC addresses, IP addresses, and port numbers) adapt during transmission, and how services operate on specific destination ports. Ultimately, observing TCP connection setups, data transfer, and session teardowns provides a clear practical mapping of theoretical OSI and TCP/IP protocol suite concepts to real-time network operations.